ISLP Chapter 9 — Support Vector Machines

Three Worked Exercises (Statements and Solutions)

Prepared for class presentation

Exercise 9.1 (Conceptual — Margins and Support Vectors) (paraphrased)

Problem. Define the geometric margin of a separating hyperplane and explain how it relates to generalization. Given β, b and training points, compute the margin and identify support vectors.

Solution. For a hyperplane $H = \{x : \beta^\top x + b = 0\}$ with $\|\beta\| > 0$, the *geometric margin* is the minimum signed distance of any training point to H:

$$\gamma = \min_i \frac{y_i(\beta^\top x_i + b)}{\|\beta\|}.$$

Points attaining the minimum are *support vectors.* A larger γ reduces the classifier's capacity (VC bound decreases), improving expected test error in separable settings.
Numerical illustration. Suppose $\beta = (2, -1)^\top$, $b = -3$ and we have positives at $(4, 4)$, $(3, 5)$ and negatives at $(1, 1)$, $(2, 0)$. Evaluating $y_i(\beta^\top x_i + b)$ gives $\{3, 2, 1, 1\}$; with $\|\beta\| = \sqrt{5}$ the margin is $\gamma = \min\{3, 2, 1, 1\}/\sqrt{5} = 1/\sqrt{5}$. The last two points are support vectors.

Exercise 9.3 (Theory — Linear Kernel Equivalence) (paraphrased)

Problem. Show that training an SVM with a linear kernel is equivalent to the maximum-margin classifier in the original feature space. State the decision function and how classification is performed.

Solution. With the linear kernel $K(x, z) = x^\top z$, the dual SVM solution is $\max_\alpha \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i\alpha_j y_i y_j x_i^\top x_j$ subject to $0 \le \alpha_i \le C$ and $\sum_i \alpha_i y_i = 0$. The primal vector is $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$, and the bias $\hat{b}$ is recovered from KKT conditions using any support vector with $0 < \alpha_i < C$. The classifier is $\hat{y} = \mathrm{sign}(\hat{\beta}^\top x + \hat{b})$. This matches the maximum-margin linear classifier because the primal objective $\frac{1}{2}\|\beta\|^2 + C\sum_i \xi_i$ and constraints are exactly the margin-maximization program with slack variables.

Exercise 9.7 (Python — SVM on Simulated Data) (paraphrased)

Problem. Simulate a two-class, nonlinearly separable dataset and fit (i) a linear SVM and (ii) an RBF SVM. Use 5-fold CV to tune C and γ for the RBF model and compare accuracy and the number of support vectors.

Solution.

```
import numpy as np
from sklearn.datasets import make_circles
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV, train_test_split
from sklearn.metrics import accuracy_score
```

```
X, y = make_circles(n_samples=1000, factor=0.5, noise=0.1, random_state=1)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, random_state=1)

lin = SVC(kernel='linear', C=1).fit(Xtr, ytr)
acc_lin = accuracy_score(yte, lin.predict(Xte))
n_sv_lin = lin.n_support_.sum()

grid = {'C':[0.1,1,10,100], 'gamma':[1,0.1,0.01,0.001]}
rbf = GridSearchCV(SVC(kernel='rbf'), grid, cv=5).fit(Xtr, ytr)
acc_rbf = accuracy_score(yte, rbf.predict(Xte))
n_sv_rbf = rbf.best_estimator_.n_support_.sum()
```

Model	Test Accuracy	# Support Vectors
Linear SVM	≈ 0.50–0.70	high
RBF SVM (tuned)	≈ 0.95–0.99	moderate

The RBF kernel captures the circular decision boundary, vastly outperforming the linear SVM and using a moderate number of support vectors.